

Locating resolution-dependent behavioral correspondence in qualitative DNA-damage response models

Carlos Ramirez Ovalle*

Department of Natural Sciences and Mathematics, Pontificia Universidad Javeriana Cali, Calle 18 No. 118-250, Via a Pance, Cali, Valle del Cauca, Colombia
carlosovalle@javerianacali.edu.co

Models with the same stable-fate predictions can differ in whether a response persists after a signal is withdrawn. We compare the published Calzone death-receptor model with its documented CASP3-to-CASP8 feedback-deletion variant, using one fixed interface observing NFkB, CASP3 and mitochondrial permeability transition. Both variants reach survival, apoptosis and necrosis under sustained TNF. An automatically extracted shared state follows 8 asynchronous updates and the visible history CASP3 activation. From that state, sustained-stimulation continuations have the same exact observable language. Allowing TNF withdrawal breaks correspondence: the feedback-positive continuation preserves CASP3 and has only an apoptotic terminal fate, whereas the feedback-negative continuation permits CASP3 deactivation and has only the model's naive terminal signature. Exact simulation certificates and an independent model checker verify this conditioned comparison. A verified 12-action counterexample also refutes global sustained trace equivalence; the reverse trace inclusion remains undecided. Thus branching alone does not separate globally trace-equivalent biological models in this case. A secondary DNA-damage example illustrates correspondence changes across three declared interfaces. The study provides a reproducible relational reconstruction of a published commitment phenomenon, not a newly discovered mechanism or wet-lab validation.

Keywords: Cell-fate commitment; death-receptor signaling; qualitative models; signal withdrawal; weak bisimulation.

1. Introduction

A cell-fate model may predict apoptosis during sustained stimulation without specifying whether that response still depends on the stimulus. This distinction matters when two mechanistic hypotheses predict the same ordinary endpoints: the informative comparison may concern the futures remaining after an intervention, rather than their original list of attractors. Death-receptor signaling provides a documented setting. Calzone et al. compared a model containing CASP3-to-CASP8 positive feedback with a variant lacking that interaction, and analyzed the persistence of apoptosis after transient versus sustained ligand exposure.¹ We use that published phenomenon to ask a narrower, executable question: *after a shared history and with*

*Corresponding author.

the same readouts, which future responses can each variant still reproduce when TNF is withdrawn?

Qualitative biological models make such hypotheses executable without requiring fully parameterized kinetic descriptions.^{2–4} Their comparison nevertheless depends on initial conditions, update semantics, and observable events. Structural similarity characterizes wiring, not the availability of actions at a reachable state.⁵ Trace comparison adds event order but can discard where alternatives cease to co-exist. Example 1 makes that mathematical distinction explicit with early and late commitment to a choice. Here it motivates the biological search for common observations followed by different intervention responses; it does not predetermine that a biological pair must have equal global traces.

We apply established weak simulation and bisimulation^{6,7} to finite executable models. In $X \preceq_w Y$, the right-hand model must match each step of the left-hand model and retain that ability at the resulting state pair. Bisimulation uses one consistent relation in both directions. The observational interface is fixed before these calculations, and hidden internal events may be matched silently. The comparison map is

$$(N_1, N_2, \Sigma) \mapsto (\mathcal{L}(N_1), \mathcal{L}(N_2)) \mapsto C_\Sigma(\mathcal{L}(N_1), \mathcal{L}(N_2)), \quad (1.1)$$

where C_Σ reports the supported relation without assigning a biological quality ranking to its classes.

The primary case compares the original death-receptor model, denoted DR-FB+, with the documented feedback-deletion variant, DR-FB-. Under sustained TNF, both have the same three reachable stable fates. We identify a shared CASP3-positive state with exactly matching sustained continuations but different responses to withdrawal. The full withdrawal-aware systems also have explicit trace counterexamples in both directions. Thus the demonstrated value is an intervention-sensitive characterization of commitment in endpoint-matched models, not exclusive superiority over trace analysis once the discriminating intervention is included. A verified finite counterexample also refutes global sustained trace equivalence, so trace analysis can distinguish the complete baseline systems as well. Only the reverse sustained trace inclusion remains undecided; no bounded trace distance is substituted for an exact decision.

A complementary GIM (genome-instability/DNA-repair) example concerns a different question: how correspondence changes when the interface itself changes. Animal/human and *Arabidopsis thaliana* encodings share broad damage-response functions but contain independently documented terminal differences.^{8–13} Their structural similarity remains 0.9976 while correspondence is lost at the first of three declared interfaces that exposes those differences. GIM is retained as an illustration of interface-conditioned correspondence, not the principal biological evidence or a search for a unique biological boundary.

The contribution is a reproducible connection between an explicit biological perturbation question, reachable future sets, and checked relational witnesses. It

recovers an already published feedback-dependent persistence phenomenon; neither the mechanism nor the idea of ligand withdrawal is a new discovery. Supporting curated cases and HPN-DREAM/CASPOTS response data retain their separate evidential roles.^{14, 15}

2. Biological comparison problem and models

2.1. Death-receptor models and the commitment question

We import the original 28-node Boolean GINsim model distributed with the Calzone study, rather than a later multiscale adaptation.¹ DR-FB+ uses its published rules. DR-FB- changes only the CASP8 rule from $(\text{DISC_TNF} \vee \text{DISC_FAS} \vee \text{CASP3}) \wedge \neg \text{cFLIP}$ to $(\text{DISC_TNF} \vee \text{DISC_FAS}) \wedge \neg \text{cFLIP}$. CASP3 remains a dynamic component; this is not its knockout. The source article’s Figures S2 and S3 explicitly examine removal of this interaction under sustained and transient stimulation. Its previously reported biological interpretation is therefore a rationale for the comparison, not independent new experimental validation of our results.

The common initial state has TNF, FADD, ATP and cIAP equal to one, and all other nodes equal to zero. FASL is zero. Each asynchronous update changes one enabled component; FASL and FADD remain fixed. The baseline protocol also fixes TNF at one. The intervention protocol additionally enables `withdraw_TNF` at every state with TNF equal to one; this action changes only TNF to zero, irreversibly, with identical availability in both variants.

The single predeclared interface observes activation and deactivation of NFkB, CASP3 and MPT, together with the withdrawal action. All other updates are silent. These are the source model’s survival, apoptotic and necrotic readouts; ATP depletion is additionally required when classifying a terminal state as necrotic. An apoptotic terminal signature has CASP3 active and NFkB/MPT inactive; survival has only NFkB active; necrosis has only MPT active and ATP depleted. The naive signature has all three markers inactive and ATP present. Mixed or varying terminal signatures are reported as other, not forced into a fate class. No such additional terminal class occurred here.

We compute reachable terminal strongly connected components and separately check whether CASP3 can ever become inactive. A singleton reachable terminal fate is not, in general, an inevitability claim under arbitrary scheduling. We use *model-level commitment under the declared withdrawal semantics* for the selected state’s unique apoptotic terminal fate together with CASP3 persistence after immediate withdrawal. The selected continuation graphs are acyclic, so their terminating post-withdrawal behavior does not require an extra fairness assumption. This does not establish cellular commitment or recovery from an executed biological death program.

The initial object-based enumeration exceeded 200,000 states. Before relation results were available, we documented an unchanged-interface implementation amendment: omit updates of the hidden sink readouts NonACD, Apoptosis and Survival,

which regulate no retained node, and enumerate retained states using compact Boolean/CSR storage. States agreeing on all retained coordinates form a weak bisimulation between the full model and this quotient: a sink update stutters, while every retained update and withdrawal lifts in either direction. All rules and initial values for the remaining components are unchanged. Weak traces, weak relations and retained terminal signatures are preserved; strong bisimilarity of a quotient is not automatically a strong claim about the unreduced model. Resource amendments, source hashes and the original configuration are retained in the reproducibility record.

2.2. *GIM as an interface-sensitivity problem*

The secondary question is whether the animal/human and *Arabidopsis thaliana* DNA-damage models encode a common executable control organization, and at which observational resolution that correspondence ceases to hold. Independent biology supports comparing broad functions but also requires preserving mechanistic distinctions. ATM- and ATR-associated sensing, checkpoint control, repair, and recovery occur in both kingdoms; mammalian p53/apoptosis and the plant-specific SOG1, WEE1, SMR, and damage-associated endoreduplication programs are not interchangeable mechanisms.^{8–13, 16, 17}

We therefore evaluated three biologically motivated interfaces on exactly the same two Petri-net structures and initial markings. Interface A observes five coarse functions: damage detection, checkpoint activation, repair, restoration, and exit from the proliferative cycle. Interface B additionally distinguishes double-strand-break and replication-stress detection, ATM- and ATR-associated signaling, and broad repair families, while retaining a common terminal `cell_cycle_exit` readout. Interface C preserves those distinctions and separately exposes mammalian `apoptosis`, plant `smr_induction`, and the plant net’s combined `differentiation_or_endoreduplication` transition. The last label is a property of this compact encoding: the model does not infer differentiation and endoreduplication as separate branches. Table 1 records why each process is observed or hidden. The executable definitions and the literature keys were fixed in source data and are exported as a machine-readable audit table.

The interfaces are curated functional abstractions, not experimentally calibrated observation operators. Their order was fixed for this comparison from known mechanisms; it does not constitute a blinded prediction. The plant and animal nets are small safe interleaving encodings, not complete DDR pathways.

Table 1. Biological rationale for the GIM observational interfaces. “A” denotes coarse function; “B/C” denotes the pathway-resolved interfaces.

Process	Animal/human encoding	<i>Arabidopsis thaliana</i> encoding	Interface treatment and independent support
Damage sensing	ATM-dominant DSB and ATR-dominant replication-stress channels	ATM/ATR channels upstream of SOG1	A merges detection; B/C distinguish the initiating channels. ^{9,10,13}
Checkpoint	CHK1/2–p53–p21 control	SOG1-regulated WEE1 and SMR control	A exposes function; B/C expose ATM/ATR but keep downstream relays internal. ^{8,11,12,16}
Repair	DSB and excision-repair families followed by recovery	Broad DSB and excision-repair families followed by recovery	A merges repair; B/C separate repair families. ^{8,10,17}
Unresolved damage	Apoptotic exit from the proliferative pool	SMR-associated arrest followed by a combined differentiation/endoreduplication outcome	A/B collapse terminal loss; C distinguishes encoded mechanisms. ^{8,12,13,18}

3. Computational comparison method

3.1. Labeled Petri nets and induced transition systems

Definition 1 (Labeled place/transition net). A labeled place/transition net is a tuple

$$N = (P, T, \text{pre}, \text{post}, \ell, m_0), \quad (3.1)$$

where P and T are finite sets of places and transitions, $\text{pre}, \text{post} : T \rightarrow \mathbb{N}^P$ are the input and output multisets, $m_0 \in \mathbb{N}^P$ is the initial marking, and $\ell : T \rightarrow \Sigma \cup \{\tau\}$ assigns either an observable action in the declared interface Σ or the internal action τ .

Under standard interleaving semantics,¹⁹ t is enabled at m when $\text{pre}(t) \leq m$ componentwise; firing it changes m to $m - \text{pre}(t) + \text{post}(t)$. The implementation uses a worklist to enumerate reachable markings explicitly. A state cap and a token bound make failure to establish a finite state space explicit rather than silently truncating the relation calculation. All curated Petri nets used in Section 2 are safe, but the definitions below apply to every finite reachable system.

Definition 2 (Finite labeled transition system). A finite labeled transition system (LTS) is

$$L = (S, \Sigma_\tau, \longrightarrow, s_0), \quad \Sigma_\tau = \Sigma \cup \{\tau\}, \quad (3.2)$$

where S is a finite state set, $s_0 \in S$, and $\longrightarrow \subseteq S \times \Sigma_\tau \times S$. We write $s \xrightarrow{a} s'$ for $(s, a, s') \in \longrightarrow$. The reachable semantics of the net in Definition 1 is denoted by

6 *C. Ramirez Ovalle*

$\mathcal{L}(N)$.

3.2. Observable and weak transitions

Definition 3 (Silent closure and weak target sets). For $s \in S$, let

$$\text{Cl}_\tau(s) = \{u \in S : s \xrightarrow{\tau^*} u\}. \quad (3.3)$$

For $a \in \Sigma$, define

$$W_a(s) = \{v : \exists u, w \quad s \xrightarrow{\tau^*} u \xrightarrow{a} w \xrightarrow{\tau^*} v\}, \quad W_\tau(s) = \text{Cl}_\tau(s). \quad (3.4)$$

We write $s \xRightarrow{a} v$ when $v \in W_a(s)$.

The convention $W_\tau = \text{Cl}_\tau$ is important: a direct silent move on one side may be matched by zero or more silent moves on the other. It is also the convention implemented by the Python engine and used in the AUT exports checked by mCRL2.

3.3. Simulation and bisimulation

Let $L_i = (S_i, \Sigma_\tau, \longrightarrow_i, s_i^0)$ for $i \in \{1, 2\}$.

Definition 4 (Weak simulation). A relation $R \subseteq S_1 \times S_2$ is a weak simulation from L_1 to L_2 if, whenever $(x, y) \in R$ and $x \xrightarrow{a}_1 x'$, there is $y' \in W_a^2(y)$ such that $(x', y') \in R$. We write

$$L_1 \preceq_w L_2 \quad (3.5)$$

if (s_1^0, s_2^0) belongs to some such relation. Thus the right-hand system L_2 simulates the left-hand system L_1 ; the left-hand behavior is contained in the right-hand behavior under the interface.

Definition 5 (Weak and strong bisimulation). A relation $R \subseteq S_1 \times S_2$ is a weak bisimulation if it satisfies the condition in Definition 4 and, for every $(x, y) \in R$ and $y \xrightarrow{a}_2 y'$, there is $x' \in W_a^1(x)$ with $(x', y') \in R$. The initial systems are weakly bisimilar, written $L_1 \sim_w L_2$, when their initial pair belongs to such a relation. Strong bisimulation, written $L_1 \sim_s L_2$, replaces every weak target set $W_a^i(x)$ by the direct target set $\{x' : x \xrightarrow{a}_i x'\}$, including for $a = \tau$.

These are standard branching-time relations.^{6,7} The explicit orientation in (3.5) prevents the common ambiguity between “simulates” and “is simulated by” in one-way results.

3.4. Weak trace language

Definition 6 (Exact and truncated weak traces). The finite weak observable language of L is

$$L_w(L) = \{a_1 \cdots a_r \in \Sigma^* : s_0 \xRightarrow{a_1} \cdots \xRightarrow{a_r} s \text{ for some } s \in S\}. \quad (3.6)$$

Its depth- k restriction is $L_{\leq k}(L) = \{u \in L_w(L) : |u| \leq k\}$. The diagnostic distance is

$$d_k(L_1, L_2) = 1 - \frac{|L_{\leq k}(L_1) \cap L_{\leq k}(L_2)|}{|L_{\leq k}(L_1) \cup L_{\leq k}(L_2)|}. \quad (3.7)$$

The implementation computes the prefix-closed truncated sets in Definition 6. The statement $d_k = 0$ is therefore not identified with exact weak-trace equivalence. Exact weak-trace results come from mCRL2, acyclic path enumeration, or finite subset-product exploration without a depth cutoff; none is inferred from a truncated distance.

3.5. Graded behavioral classification

Definition 7 (Graded classifier). For finite L_1, L_2 over the declared interface Σ , define $C_\Sigma(L_1, L_2)$ by the following priority:

$$C_\Sigma(L_1, L_2) = \begin{cases} \text{strong,} & L_1 \sim_s L_2, \\ \text{weak,} & L_1 \not\sim_s L_2 \text{ and } L_1 \sim_w L_2, \\ \text{mutual,} & L_1 \not\sim_w L_2, L_1 \preceq_w L_2, L_2 \preceq_w L_1, \\ \text{left_in_right,} & L_1 \preceq_w L_2, L_2 \not\preceq_w L_1, \\ \text{right_in_left,} & L_2 \preceq_w L_1, L_1 \not\preceq_w L_2, \\ \text{noncomparable,} & \text{otherwise.} \end{cases} \quad (3.8)$$

The output is a reporting class, not a newly proposed equivalence relation.

Proposition 1 (Exclusive reporting). *The classifier in Definition 7 assigns exactly one class to every ordered pair of finite LTSs.*

Proof. The first two cases are selected before any simulation-only case. If neither is selected, the two simulation predicates have exactly four Boolean combinations: both true, only the first true, only the second true, or both false. These are the last four cases of (3.8), respectively. The cases are exhaustive and disjoint by their stated guards. $\square$

3.6. Computational implementation

Algorithm 1 summarizes the executable path. Reachability and the formal relations are exact whenever the finite state space is completed below the declared state and token caps. The distance d_k , PN-GDDA, LTS-GDA, and the structural profile are diagnostics rather than formal relations.

Algorithm 1 (Executable comparison pipeline).

- (i) Enumerate the reachable LTS from the initial marking or logical state by a worklist; stop with an explicit error if a cap is exceeded.

- (ii) Cache $\text{Cl}_\tau(s)$ and construct $W_a(s)$ by collecting a -successors from the closure and the closures of their targets.
- (iii) Initialize $R = S_1 \times S_2$ and delete a pair immediately when it fails the selected transfer operator; repeat complete passes until none is deleted.
- (iv) Evaluate strong and weak bisimilarity and both simulation directions, apply Eq. (3.8), and attach trace and structural diagnostics.

The finite relation-deletion theorem and the strict hierarchy underlying this reporting priority are proved in Appendix A. Example 1 provides an explicit simulation witness and a failed-successor proof for equal traces with one-way simulation. PN-GDDA, LTS-GDA and trace distances answer complementary diagnostic questions; their definitions and implementation checks are retained in Supplementary Validation, Section S1.

The accepted early/late-choice example supplies the conceptual target: equal histories need not preserve the same future choices. The biological analysis below tests a related intervention question without assuming equal global trace languages.

4. Main result: signal withdrawal exposes a commitment difference

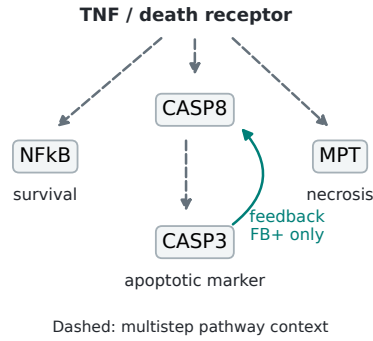
4.1. Baseline agreement and its exact scope

Complete reachable-state enumeration yields 1,044,457 states and 547,021 states for the two sustained-stimulation quotients. Each has exactly three reachable singleton terminal components: survival, apoptosis and necrosis. This is exact agreement of terminal-fate predictions, not an assertion about matching probabilities or every transient trajectory. An expanded exact subset-product search found a 12-action word admitted by DR-FB+ but not DR-FB-, without changing the models or interface:

$$w = \text{MPT}\uparrow, \text{CASP3}\uparrow, \text{CASP3}\downarrow, \text{NFkB}\uparrow, \\ \text{NFkB}\downarrow, \text{MPT}\downarrow, (\text{CASP3}\uparrow, \text{CASP3}\downarrow)^3.$$

The superscript denotes three repetitions of the action pair, not physical time. Exact propagation, an independent sparse reachability calculation and mCRL2 confirm acceptance only in DR-FB+. A reconstructed 53-update accepting path also satisfies the original full 28-node rules. Therefore global sustained trace equality is refuted, DR-FB- cannot weakly simulate DR-FB+, and the two systems are not weakly bisimilar. The reverse inclusion remains unresolved; its unknown status does not prevent the inequality decision. This is a qualitative model counterexample, not a measured oscillatory trajectory.

The more specific comparison at the shared state identified below is fully resolved. With TNF kept on, each continuation has 3 states and 2 silent edges, and no further observable action. Both are strongly and weakly bisimilar and have exact weak language $\{\epsilon\}$. The equality concerns continuations from that state, not the complete systems from their original physiological initial condition.

A Published mechanistic contrast**B Sustained TNF: same stable fates**

	FB+	FB-
Survival	●	●
Apoptosis	●	●
Necrosis	●	●

Exact terminal-fate agreement
Global sustained traces differ (verified word)

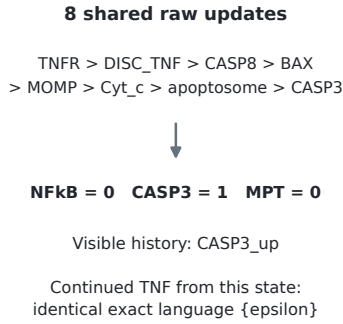
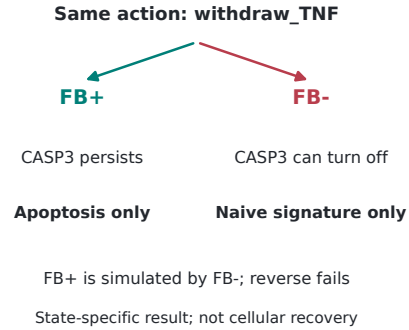
C Shared state, fixed observations**D Withdraw TNF from that state**

Fig. 1. Fixed-interface commitment comparison. (A) Mechanistic context, with the literature-described CASP3-to-CASP8 feedback highlighted; dashed connectors summarize multistep pathways, not direct molecular interactions. (B) Both sustained-stimulation models reach the same three stable fates. (C) An automatically extracted common history reaches the same retained state; with continued TNF, both continuations have language $\{\epsilon\}$. (D) Withdrawal from that state yields persistent CASP3 and an apoptotic terminal fate for DR-FB+, but permits CASP3 loss and only the naive terminal signature for DR-FB-. Counts and relations refer to exact retained-state executions. No event count is converted to physical time.

4.2. A common history and a state-specific commitment witness

The deterministic breadth-first search selects the first shared reachable state satisfying persistent CASP3 and a unique apoptotic terminal future after withdrawal in DR-FB+, versus a unique naive terminal future in DR-FB-. One shortest raw-update path to this state is

TNFR↑, DISC_TNF↑, CASP8↑, BAX↑, MOMP↑,
Cyt_c↑, apoptosome↑, CASP3↑.

All eight updates and the resulting retained vector are available in both variants. The visible projection is $h = \text{CASP3_up}$. At this state, TNF, FADD, ATP, cIAP, TNFR, DISC_TNF, CASP8, BAX, MOMP, Cyt_c, apoptosome and CASP3 are one; all other retained components are zero. Thus the selected state is fully specified, not merely a pair of matching marker values. The automatic selection is minimal in raw-event depth among the states satisfying this criterion; it is not a claim of a globally shortest biological observation or of a unique commitment state.

When withdrawal is permitted from this state, the continuations have 12 states/17 edges and 34 states/54 edges, respectively. Write them K_+ and K_- . Exact checks give

$$K_+ \preceq_w K_-, \quad K_- \not\preceq_w K_+, \quad K_+ \not\preceq_w K_-.$$

The distinguishing trace in K_- is **withdraw.TNF CASP3_down**, absent from K_+ . The independent deletion-game certificate covers every defending weak reply, not one conveniently chosen successor. Its dependency ranks decrease until CASP3 deactivation has no matching action. Production predicates and mCRL2 agree on bisimulation and exact traces; both simulation directions are also checked by the independent game.

The direction is important: K_- can match the fewer observable actions of K_+ , but K_+ cannot match the additional deactivation response of K_- . This is not a ranking of biological quality or survival. After immediate withdrawal, K_+ retains CASP3 along every continuation and reaches only apoptosis, whereas K_- reaches only the model's naive signature. The latter is a marker-level logical outcome, not evidence that an apoptotic cell can return to viability.

4.3. *What the shared observation does and does not identify*

The visible history h alone is compatible with 78 hidden states in each sustained model. Aggregating complete post-withdrawal futures over all such states gives

$$\begin{aligned} \text{Futures}_+(h, \text{withdraw}) &= \{\text{apoptosis, naive, necrosis}\}, \\ \text{Futures}_-(h, \text{withdraw}) &= \{\text{naive, necrosis}\}. \end{aligned}$$

Therefore the difference is not confined to an arbitrarily paired pair of hidden states. However, h does not imply that every compatible DR-FB+ state is irreversibly apoptotic. The full state witness and the history-aggregated future sets answer different questions and are reported separately.

The complete withdrawal-aware systems have 3,133,393 and 2,140,569 reachable states. Exact subset propagation gives a DR-FB-only trace **CASP3_up withdraw.TNF CASP3_down**; the defending DR-FB+ subset is empty after its last action. A trace in the opposite direction is **withdraw.TNF CASP3_up CASP3_down CASP3_up**. These exact counterexamples refute both global trace inclusions and hence both global weak-simulation directions; they require no exhaustive language enumeration. Ordinary trace comparison can detect differences under both the sustained and withdrawal protocols.

Table 2. Specified assumptions, computed results, and absent evidence in the death-receptor comparison.

Item	Evidential status
CASP3-to-CASP8 feedback	Published model assumption; its removal was already studied.
Markers and withdrawal	Fixed interface and intervention declared before relation results.
Baseline terminal fates	Computed agreement: survival, apoptosis, necrosis.
Shared-state continuations	Exact sustained equality; one-way simulation and unequal traces with withdrawal.
Global sustained traces	Equality refuted by a verified 12-action word; reverse inclusion undecided.
Witness and fate persistence	Automatically computed; all defending replies checked.
Discriminating observation	Model-derived implication of a published perturbation.
Wet-lab confirmation	No new experimental validation; naive is not demonstrated cellular recovery.

We do not present this biological case as the stronger equal-global-trace/different-branching phenomenon of Example 1.

The exhaustive scan considers every reachable history of exactly k raw updates before withdrawal, for $0 \leq k \leq 40$, with complete future reachability thereafter. An apoptotic terminal future first becomes possible in DR-FB+ after one hidden TNFR update; persistent-CASP3 commitment first appears at $k = 8$, for 1 of 73 reachable states at that depth. It never occurs in DR-FB- over this scan. State counts are not path probabilities. No artificial stutters are added at terminal states, no physical duration is assigned, and this first possible depth is not a universal time at which all modeled cells commit.

4.4. Prospective discriminator and evidential status

The model-derived discriminator is a pulse/withdrawal comparison at a CASP3-positive stage, followed by the joint observation of CASP3 persistence, NFkB activity and MPT, with ATP status supporting fate classification. Monitoring the other two markers matters: CASP3 loss accompanied by a change in the necrotic branch is not the same trace as isolated CASP3 loss after withdrawal. The three-marker history does not uniquely locate the hidden witness state; more detailed state assessment would be needed to test its state-specific claim. We specify no concentration, cell line, elapsed time or selective experimental feedback inhibitor that these logical models cannot justify. Ligand withdrawal and the feedback hypothesis originate in the Calzone study.¹ Our added result is the explicit relational and reachable-future account, not a newly proposed biological mechanism.

5. GIM illustration: interface-conditioned behavioral correspondence

5.1. *Structure held constant*

The animal and plant nets reach 13 and 14 states, respectively. No place, transition, arc, or initial marking changes across the three interfaces. Native PN-GDDA is 0.9975846 throughout (0.9975175 with the independently derived 576-orbit partition). This describes nearly identical selected local wiring, not mechanistic identity or execution equivalence. Figure 2 makes the comparison’s biological assumptions and computed outcomes visible together.

5.2. *Coarse function retains correspondence*

At Interface A, every observable detection, checkpoint, repair, restoration, or cycle-exit step in either executable model can be matched by the other, with this matching preserved at the resulting states. Internal relays can require different numbers of steps. Formally, the systems are weakly but not strongly bisimilar. The result licenses comparison at the declared shared functional level; it does not identify their molecular implementations.

5.3. *Pathway distinctions do not yet break the relation*

Interface B distinguishes damage channels, ATM/ATR-associated signaling, and broad repair families. The models still match recursively when terminal loss of proliferative capacity remains a common observation. Both simulation directions and weak bisimilarity hold; strong bisimilarity does not. Thus, among these encodings and interfaces, distinguishing those pathway events alone is insufficient to eliminate the correspondence.

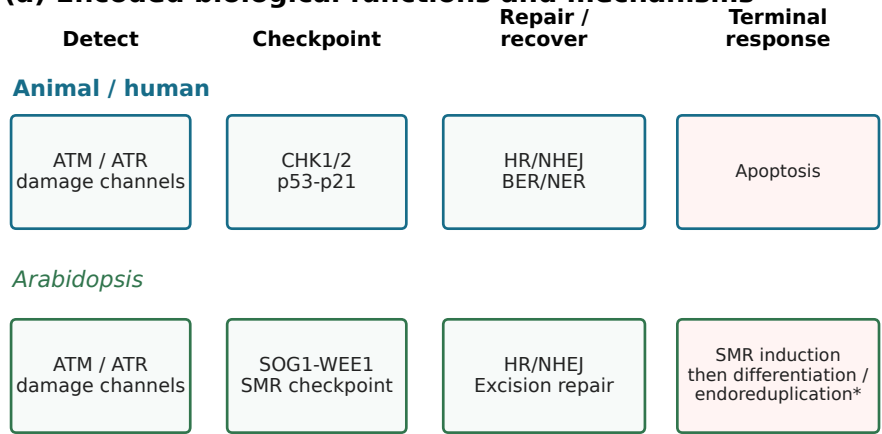
5.4. *Terminal mechanisms eliminate both directions*

At Interface C, mammalian `apoptosis` and plant `smr_induction` followed by `differentiation_or_endoreduplication` become distinct observations. Neither model can match all observable moves of the other, so weak bisimilarity and both simulation directions fail. This is not an unexpected biological discovery: the terminal labels intentionally encode prior knowledge of different mechanisms. The computational result confirms where, in the tested sequence of readouts, that distinction prevents recursive matching. The depth-six trace distance becomes 0.3529 rather than zero; LTS-GDA changes from 0.9508 to 0.8921. These diagnostics accompany the formal decision, not replace it (Table 3).

5.5. *What changes when observability changes?*

Interfaces A and B aggregate terminal mechanisms into a shared functional readout, whereas C separates them. The first loss of correspondence is therefore *between*

(a) Encoded biological functions and mechanisms



*One combined terminal transition in the plant encoding, not two alternatives.

(b) Same structures, different observational resolution

Readout	What is distinguished	PN-GDDA	Branching relation
A Coarse	Shared functions; terminal loss of proliferative capacity	0.9976	Weakly bisimilar
B Pathway	Damage, ATM/ATR and repair; terminal mechanisms collapsed	0.9976	Weakly bisimilar
C Terminal	Pathway events plus separate animal / plant terminal mechanisms	0.9976	Non-comparable (both directions fail)

Model- and interface-conditional results; no organism-level equivalence

Fig. 2. Central GIM comparison. Upper-panel columns align encoded animal/human and *Arabidopsis thaliana* functional categories; they are not a causal pathway or a required sequence of outcomes. The lower panel shows what each declared interface observes and the computed branching relation. PN-GDDA remains 0.9976 because the structures are unchanged. The combined plant terminal transition does not distinguish differentiation from endoreduplication. Shared labels at A/B express a functional abstraction, not identity between p53 and SOG1 or between terminal mechanisms.

B and *C* within this declared interface family, not a uniquely identified or globally minimal biological resolution. There is no exhaustive search over all possible interfaces, and no experimental threshold is inferred. The constant structural score isolates observability from topological change.

Prior biology supplies the distinct terminal mechanisms; PN-GDDA describes their selected wiring; the executable comparison determines whether a recursive cor-

Table 3. Recomputed GIM results. $\text{Animal} \preceq \text{Plant}$ means that the plant model simulates the animal model; $\text{Plant} \preceq \text{Animal}$ is the reverse. Strong bisimilarity is false throughout.

Interface	$\text{Animal} \preceq \text{Plant}$	$\text{Plant} \preceq \text{Animal}$	Class	d_6	PN-GDDA
A: coarse	Y	Y	weak bisimulation	0.0000	0.9976
B: pathway	Y	Y	weak bisimulation	0.0000	0.9976
C: terminal	N	N	non-comparable	0.3529	0.9976

Table 4. Supporting curated model-pair outputs. GIM is reported at Interface A; all relations and d_6 values are interface-conditional, whereas PN-GDDA is structural.

Module	Process	Formal class	d_6	PN-592	Orbit-576
GIM	DNA repair	weak bisimulation	0.000	0.9976	0.9975
DCE	energy metabolism	weak bisimulation	0.000	0.9969	0.9968
SPS	cell-cycle control	weak bisimulation	0.000	1.0000	1.0000
RCD	death/autophagy	plant in animal	0.143	1.0000	1.0000
AID	immune control	non-comparable	0.429	0.9907	0.9904

response survives each declared readout. Whether such a model-level boundary predicts a cross-species experimental boundary remains unvalidated. One testable interpretation is that readouts aggregating terminal loss of proliferative capacity may preserve an apparent correspondence that mechanism-specific readouts do not. This is a model-derived hypothesis, not an assay result. The full Petri nets remain in Appendix B.

6. Supporting and exploratory comparisons

Five compact plant–animal pairs encode DNA repair/genome instability (GIM), energy metabolism (DCE), cell-cycle control (SPS), cell death/autophagy (RCD), and immune control (AID). Comparative studies and cancer-hallmark literature guided component selection; dedicated reviews informed plant DNA-damage, cell-cycle, and autophagy mechanisms.^{10, 20–22} Transition-level provenance is machine-checked, which establishes where encoded assumptions came from and whether documentation is complete. It does not establish that the nets reproduce independent experiments. These manually curated models contain no kinetics, concentrations, tissue context, or independent experimental calibration.

The ten nets produced 5–14 reachable states. The GIM row in Table 4 reports Interface A; GIM, DCE, and SPS were weakly but not strongly bisimilar; RCD gave plant-in-animal one-way simulation; AID was non-comparable. Table 4 juxtaposes these classes with native PN-GDDA. The 592- and 576-orbit variants changed no score by 0.001 and no threshold decision: all five pairs were structurally similar at 0.9 despite three different behavioral classes.

RCD supplies a secondary directional result: both encoded nets contain stress, mTOR inhibition, autophagy, and survival, but only the animal encoding exposes a

canonical caspase-apoptosis branch; the plant branch uses an internal metacaspase event. The plant LTS is weakly simulated by the animal LTS, whereas the reverse direction fails. This means containment only for the declared labels of these two models; it does not equate plant programmed cell death with mammalian apoptosis. Silent subdivision preserved every class over 300 refinements per module, while targeted relabeling changed the expected relations. Distances against 2,000 label permutations assess internal interface sensitivity, not external biological significance. DCE, SPS, and AID are retained as supporting model-ingestion and relation-diversity examples rather than independent biological validation.

6.1. *Exploratory confrontation with HPN-DREAM data*

An independent-data check is retained as a secondary, exploratory analysis, not as validation of the GIM resolution boundary. Public HPN-DREAM/CASPOTS families contain 72 BT20, 191 BT549, and 21 MCF7 Boolean networks. Structure-only medoid selection, fixed before test-response access, chooses zero-based rows 24, 137, and 9.^{14,15} The common interface contains seven measured PI3K/MAPK/mTOR readouts under three shared held-out mTOR-inhibitor conditions. Initial values, clamping, and update conventions are specified in Supplementary Validation, Section S3.

The nine asynchronous pair-condition comparisons contain six one-way simulations and three non-comparable pairs. The six directions comprise three left-model-in-right-model and three right-model-in-left-model relations, not positions on an ordinal similarity scale. Direction is preserved in the result fields; no scalar formal-class test is performed. Median profile RMSE is 0.2757, 0.1919, and 0.2071 in those three nominal categories, respectively, with only three observations per category. Synchronous updating preserves seven of the nine classes. These summaries cannot establish predictive value.

The retained continuous diagnostics are inconclusive: truncated-trace distance has Spearman $\rho = -0.596$, exact one-sided $p = 0.556$; LTS-GDA distance has $\rho = 0.669$, $p = 0.097$, using all 216 condition-stratified permutations. The CASPOTS native-context advantage over cross-cell medoids is 0.00212 excess-RMSE units and is not established by the exact three-cell sign test ($p = 0.25$). No more favorable statistic was substituted. The full methods, negative results, and plot remain in Supplementary Validation, Section S3; they show a route to data confrontation without claiming prognosis, native cell-line specificity, or experimental equivalence.

7. Supporting formal and implementation checks

Definitions and the executed relation-deletion method are supported by the proofs in Appendix A. Table 5 states the checks, independent bases, and observed outcomes in the main article. These checks address whether the implemented relations and structural score behave as specified; they do not establish biological validity.

Table 5. Main implementation-verification results. Agreement refers to the declared inputs and predicates, not predictive accuracy on biological data.

Check	Independent basis and scope	Observed result
Constructed cases	Six pairs with classes fixed before classification	6/6 classes and 6/6 binary weak-equivalence decisions recovered.
External equivalence	mCRL2 on six synthetic, six public-model controls, and nine HPN pairs	42/42 strong/weak-bisimilarity decisions agree.
Directional audit	Raw-edge attacker–defender game on 32 model pairs and three hierarchy witnesses	No disagreement on either weak-simulation direction or strong/weak bisimilarity.
Exhaustive finite check	All 260 one-/two-state LTSs over $\{a, \tau\}$; 67,600 ordered pairs	No simulation or weak-bisimulation discrepancy; no hierarchy or trace-inclusion violation.
Structural score	Holmes 1.1.1: 151 graphlets, 592 slots; four-node net and one-arc-deletion control	PN-GDDA agrees to 12 decimal places; absolute difference 3.33×10^{-16} .

The public cell-cycle and p53–Mdm2 controls test SBML-qual and GINML imports; all six copy, silent-refinement, and label-perturbation controls recover their predeclared outcomes.^{23–26} Holmes supplies an external reference for the PN-GDDA implementation.²⁷ mCRL2 supplies the external equivalence checks.²⁸ Its simulation-preorder checks are used only for the two tau-free hierarchy witnesses, where strong and weak simulation coincide; no general weak-preorder verification is attributed to that tool.

For directional checks, the independent game constructs weak targets directly from raw edges without calling the production closure or relation routines. The 32 pairs comprise six synthetic, three GIM-interface, five curated, and 18 asynchronous/synchronous HPN comparisons. The exhaustive universe fixes initial state zero and enumerates every edge subset; it is not a proof for arbitrary finite systems. In particular, Example 1 separately establishes early-to-late simulation, failure of the reverse, and exact trace equality using an explicit witness and a failed-successor argument. No exact trace conclusion is inferred from a truncated distance.

The evidence layers remain distinct: proofs establish mathematical claims; tests establish implementation agreement on tested inputs; model provenance and import checks establish executable assumptions; biological literature and held-out data support or limit biological interpretation. Formal agreement alone does not calibrate a curated net against experiments. Supplementary Validation S1–S2 and S4 retain per-case outputs, comparator definitions, tool procedures, and scaling details; the main death-receptor evidence is in Section 4 and GIM in Section 5.

The new death-receptor checks distinguish global from conditioned comparisons. Compact Boolean execution is checked against the tuple-based executor; the selected full 28-node continuations reproduce the quotient’s weak results. The exact conditioned sustained languages agree, while withdrawal gives one-way simulation, confirmed by a raw-edge game certificate covering every defender reply. mCRL2 independently confirms all six strong/weak/trace equivalence decisions across the two conditioned protocols. Global withdrawal simulations are refuted by exact trace counterexamples in both directions. An independent tuple/product search using all 28 original rule callbacks confirms acceptance in one variant and exhaustive rejection in the other for each withdrawal counterexample, without compact storage or sink reduction. Global sustained trace equality is also refuted: a 12-action word is possible only in DR-FB+, as independently checked by sparse reachability and mCRL2. A reconstructed 53-update accepting path satisfies the original full-model rules. The reverse sustained inclusion remains undecided. The expanded subset search still reaches its storage cap, but the finite witness suffices to disprove equality; this is not an inference from resource exhaustion. Supplementary Validation S5 gives the reduction argument, full graph sizes, execution limits and event-depth scan. These limits are not counted as failed equivalences or as successful biological validations.

8. Discussion

8.1. *Can commitment matter when ordinary predictions agree?*

The death-receptor case gives a qualified but concrete answer. The two variants share their sustained stable-fate set. At an automatically located shared state, even their complete sustained observable continuations agree, yet ligand withdrawal exposes different persistent marker behavior and terminal futures. This separates end-point agreement from intervention-conditioned commitment using a fixed biological interface. The analysis connects a literature-derived mechanism to an explicit history, retained state, and recursive-matching certificate instead of relying on re-labeled terminal mechanisms.

The scope must remain precise. Global sustained weak-trace equivalence is refuted by a verified finite word, and the withdrawal-aware languages are also unequal. The reverse sustained trace inclusion remains undecided. The biological example consequently does not show that bisimulation discovers a difference invisible to every trace-based comparison. Its value is the systematic characterization of where matching fails and which futures remain available under an intervention. The exact equal-trace/unequal-branching phenomenon remains established by the accepted synthetic example; a biological instantiation of that stronger property is still an open extension.

Nor do the markers uniquely identify a hidden commitment state. Although the same visible prefix has different aggregated future sets in the two models, it is compatible with 78 internal states in each. Presenting one state-specific certificate

as an unconditional prediction for every CASP3-positive cell would erase the very nondeterminism the comparison is intended to respect. The model’s naive terminal signature after CASP3 loss must also not be read as actual reversal of executed apoptosis. Downstream irreversible damage, cell-type specificity and quantitative kinetics are not established by this qualitative reconstruction.

Calzone et al. already analyzed feedback-dependent persistence under transient ligand exposure.¹ Recovering that result is a use-case validation of the relational workflow, not independent discovery or experimental confirmation. The asymmetric simulation result adds an explicit account of which continuation fails to match; the fate analysis supplies its biological interpretation. These are complementary outputs, not interchangeable measures.

8.2. *Can changing the interface change correspondence?*

GIM addresses this separate methodological question. PN-GDDA characterizes label-blind local Petri-net topology,⁵ whereas the executable relations depend on labels, initial markings, and reachable choices. For the unchanged GIM nets, weak correspondence persists at Interfaces A and B and is lost at C, the first tested interface exposing the known terminal distinction. This result is limited to those three declared readouts; there is no exhaustive interface search, unique biological boundary, or globally minimal resolution claim. The distinct terminal biology was encoded in advance, not discovered by formal comparison.

The two cases must not be conflated. GIM changes observability while holding the models fixed. The death-receptor study holds the interface fixed and compares a documented mechanistic variant under sustained and withdrawal protocols. PN-GDDA is retained for the former structural question; no uncomputed PN-GDDA score is assigned to the death-receptor models.

8.3. *Validation, execution assumptions and limits*

The accepted mathematical development, direction conventions, and fixed-point algorithm are unchanged. Formal regression tests establish computational correctness under declared semantics, not biological adequacy. The new analysis adds independent compact/tuple execution checks, a checked hidden-sink reduction, full-model verification of the selected continuation, exact subset counterexamples, and mCRL2 checks of the local equivalences and the sustained counterexample word. State-space and memory limits remain visible results, not negative relation decisions.

Unitary asynchronous updates encode possible orders, not calibrated rates. Reachable terminal fates and marker persistence are distinguished from inevitability under unrestricted scheduling. The selected post-withdrawal continuations are acyclic, but no general fairness conclusion is inferred for the much larger global graphs. Resource amendments change storage and remove only proven irrelevant hidden readouts; they do not retune the biological interface after inspecting relation outcomes. The prospective configuration record is local, not an externally

registered protocol or proof of absence of analytic bias.

The HPN-DREAM/CASPOTS confrontation remains exploratory and its weak evidence does not become stronger because the death-receptor example was added. Likewise, the compact curated networks are not full organism-level pathways. Future work should test other independently authored commitment models, resolve the global sustained relations where feasible, and use state-resolved experimental observations to challenge the predicted intervention responses.

9. Conclusion

Endpoint-matched qualitative cell-fate models can differ in their responses to signal withdrawal. With a fixed interface, the death-receptor feedback variants share a reachable CASP3-positive state with exactly equivalent sustained continuations but unequal withdrawal continuations and different terminal futures. An automatic certificate explains the failed matching action and preserves the simulation direction. This is a model-level reconstruction of a published commitment phenomenon, not a new mechanism, proof of global trace-equivalent biological branching, or wet-lab validation. GIM remains a complementary illustration of the first correspondence loss among three declared interfaces. Together, the cases separate what is specified through models and readouts from what relational and reachable-future calculations actually establish.

Appendix A. Mathematical properties and quantifier audit

For $a \in \Sigma_\tau$, let $D_a^i(x) = \{x' : x \xrightarrow{a}_i x'\}$ and $U = S_1 \times S_2$. Define operators on $\mathcal{P}(U)$ by

$$\Phi_{\leq}(R) = \{(x, y) \in U : \forall a, x' \in D_a^1(x) \exists y' \in W_a^2(y), (x', y') \in R\}, \quad (\text{A.1})$$

$$\Phi_w(R) = \{(x, y) \in \Phi_{\leq}(R) : \forall a, y' \in D_a^2(y) \exists x' \in W_a^1(x), (x', y') \in R\}, \quad (\text{A.2})$$

$$\begin{aligned} \Phi_s(R) = \{(x, y) \in U : \forall a, x' \in D_a^1(x) \exists y' \in D_a^2(y), (x', y') \in R, \\ \forall a, y' \in D_a^2(y) \exists x' \in D_a^1(x), (x', y') \in R\}. \end{aligned} \quad (\text{A.3})$$

Theorem 1 (Correctness of relation deletion). *For each $\Phi \in \{\Phi_{\leq}, \Phi_w, \Phi_s\}$, the operator is monotone. The implemented algorithm starts with $R = U$, repeatedly deletes any currently examined pair $z \notin \Phi(R)$, and stops after a complete pass with no deletion. On finite LTSs it terminates and returns the greatest fixed point $\nu\Phi$. Consequently, testing membership of (s_1^0, s_2^0) computes, respectively, weak simulation in the orientation of Definition 4, weak bisimilarity, and strong bisimilarity.*

Proof. If $R \subseteq Q$, every witness pair in R remains available in Q ; hence $\Phi(R) \subseteq \Phi(Q)$ for each operator. Every deletion strictly decreases the finite relation, so termination requires at most $|U|$ deletions.

Let $G = \nu\Phi$. The invariant $G \subseteq R$ holds initially. If it holds before a deletion and $z \in G$, then $z \in \Phi(G) \subseteq \Phi(R)$, so z cannot be deleted. Thus G remains a

Table 6. Distinct witnesses for the three failed converses. All pairs have equal exact weak-trace languages; $X \preceq Y$ means Y simulates X .

Converse refuted	Witness pair (X, Y)	Decisive properties
Weak implies strong bisimilarity	$a.0, a.\tau.0$	weak, not strong
Mutual simulation implies weak bisimilarity	$a.(b + c) + a.b, a.(b + c)$	both directions, not weak
Exact trace equality implies mutual simulation	late choice, early choice	early $\preceq$ late only

subset of the terminal relation R_* . A final pass without deletion gives $R_* \subseteq \Phi(R_*)$. Conversely, each $z \notin R_*$ was deleted from some R_z with $z \notin \Phi(R_z)$. Since $R_* \subseteq R_z$, monotonicity implies $\Phi(R_*) \subseteq \Phi(R_z)$ and therefore $z \notin \Phi(R_*)$. Hence $R_* = \Phi(R_*)$. Because R_* is a fixed point, $R_* \subseteq G$; together with the invariant this yields $R_* = G$. The three operators are exactly the transfer clauses of Definitions 4 and 5. $\square$

Remark 1 (Resource bound). Writing $n_i = |S_i|$, the candidate relation stores $O(n_1 n_2)$ pairs and permits at most $n_1 n_2$ deletions and $n_1 n_2 + 1$ passes. Each pass scans current pairs, outgoing moves, and matching direct or weak targets. Silent closures are cached; visible weak targets are constructed on demand. We therefore report these parameters instead of claiming an unjustified tight time bound.

Proposition 2 (Hierarchy of behavioral relations). *For finite LTSs under Definitions 3–6,*

$$L_1 \sim_s L_2 \implies L_1 \sim_w L_2 \implies (L_1 \preceq_w L_2 \wedge L_2 \preceq_w L_1) \implies L_w(L_1) = L_w(L_2). \quad (\text{A.4})$$

The converses do not hold in general.

Proof. A direct match is a permitted weak match, and each direction of a weak bisimulation is a weak simulation. If $L_1 \preceq_w L_2$, induction along any finite concrete path matches observable moves by $\tau^* a \tau^*$ and silent moves by τ^* , preserving related successors. Erasing τ gives $L_w(L_1) \subseteq L_w(L_2)$; the reverse simulation gives equality. Proposition 3 and Example 1 refute the first and last converses. For the middle converse, compare $a.(b + c) + a.b$ with $a.(b + c)$: each simulates the other, but the $a.b$ successor cannot be bisimilar to a state offering both b and c . $\square$

For the middle row, write x_{bc}, x_b for the two a -successors of X and y_{bc} for the sole a -successor of Y . A simulation from X to Y relates both x_{bc} and x_b to y_{bc} , with corresponding terminal pairs. A simulation from Y to X relates y_{bc} only to x_{bc} . Both initial pairs therefore survive. A bisimulation would additionally need to match X 's transition to x_b with y_{bc} ; the latter's c -move has no counterpart at x_b . This witness concerns only the middle converse, not the trace-equality converse refuted by Example 1.

Proposition 3 (Controlled silent edge refinement). *Let L contain $p \xrightarrow{a} q$ with*

$a \in \Sigma$ and $p \neq q$. Construct L' by copying all original states and transitions except this edge, and replacing it by

$$\bar{p} \xrightarrow{a} r \xrightarrow{\tau} \bar{q}, \quad (\text{A.5})$$

where r is fresh and has no other incident behavior. Then $L \sim_w L'$. Repeated refinements of this form also preserve weak bisimilarity. If p and $\bar{p}$ are initial, the selected edge is the only a -edge from $\bar{p}$ in L' , and every a -successor of p in L has no outgoing τ transition, then $L \not\sim_s L'$.

Proof. Let $R = \{(x, \bar{x}) : x \in S\} \cup \{(q, r)\}$. For every copied pair other than the source, direct moves are copied. At $(p, \bar{p})$, the original edge is matched by $\bar{p} \xrightarrow{a} r \xrightarrow{\tau} \bar{q}$, while $\bar{p} \xrightarrow{a} r$ is matched by $p \xrightarrow{a} q$ and ends in (q, r) . At (q, r) , moves from q are matched after $r \xrightarrow{\tau} \bar{q}$; that silent move is itself matched by zero silent moves at q . Hence R is a weak bisimulation, and repetition follows by transitivity. Under the additional conditions, a strong match for $\bar{p} \xrightarrow{a} r$ must relate r to an a -successor x of p , but $r \xrightarrow{\tau} \bar{q}$ would require the excluded direct τ -move from x . Thus strong bisimilarity fails. $\square$

Proposition 4 (Limit of label-blind structural comparison). *Let M be any comparator determined only by unlabeled structure and invariant under relabeling of transitions. Then M cannot characterize weak-trace equality, weak simulation or weak bisimilarity for labeled systems in general.*

Proof. Take two systems with the same two states and one directed edge. Label the edge a in L_a and $b \neq a$ in L_b . Their unlabeled structures are identical, so $M(L_a, L_b) = M(L_a, L_a)$. Yet their languages are $\{\epsilon, a\}$ and $\{\epsilon, b\}$, and neither observable move can be matched by the other system. Thus no decision based only on M can characterize the stated labeled relations. $\square$

Proposition 4 does not identify a defect in PN-GDDA. It states that a label-blind structural method answers a different question and cannot substitute for a labeled behavioral relation.

Example 1 (Equal traces, different branching). Let the late-choice system start at ℓ_0 and have exactly

$$\ell_0 \xrightarrow{a} \ell_1, \quad \ell_1 \xrightarrow{b} \ell_b, \quad \ell_1 \xrightarrow{c} \ell_c, \quad (\text{A.6})$$

and let the early-choice system start at e_0 and have exactly

$$e_0 \xrightarrow{a} e_b, \quad e_0 \xrightarrow{a} e_c, \quad e_b \xrightarrow{b} e'_b, \quad e_c \xrightarrow{c} e'_c. \quad (\text{A.7})$$

All unlisted states are terminal; neither system has silent transitions or cycles. Enumerating all paths, whose lengths are at most two, gives both exact prefix-closed languages as $\{\epsilon, a, ab, ac\}$, not merely a truncated match.

An explicit simulation from early to late is

$$R_{E \rightarrow L} = \{(e_0, \ell_0), (e_b, \ell_1), (e_c, \ell_1), (e'_b, \ell_b), (e'_c, \ell_c)\}. \quad (\text{A.8})$$

Both initial a -moves are matched by $\ell_0 \xrightarrow{a} \ell_1$. At (e_b, ℓ_1) the sole b -move has a match; at (e_c, ℓ_1) the sole c -move has a match. Terminal pairs impose no further obligations. Thus $L_{\text{early}} \preceq_w L_{\text{late}}$.

Conversely, suppose a simulation R from late to early contains (ℓ_0, e_0) . Matching the single transition $\ell_0 \xrightarrow{a} \ell_1$ requires choosing one successor of e_0 and including the resulting pair in R . If $(\ell_1, e_b) \in R$, the c -move from ℓ_1 has no match. If $(\ell_1, e_c) \in R$, its b -move has no match. These exhaust the possible a -successors, so no such R exists: $L_{\text{late}} \not\preceq_w L_{\text{early}}$. Weak bisimilarity therefore also fails.

Different outgoing transitions from a related pair may indeed choose different matching successors. However, the *one* initial late a -transition must be matched by an early successor pair that itself satisfies every subsequent transfer obligation. Trace matching may select the e_b branch for ab and the e_c branch for ac independently; simulation cannot combine those two states into one defender state after the initial move. This is the quantifier distinction behind the final strictness claim of Proposition 2.

Appendix B. Detailed GIM Petri-net encodings

Figures 3 and 4 show the fixed executable nets; the three interfaces change only transition labels.

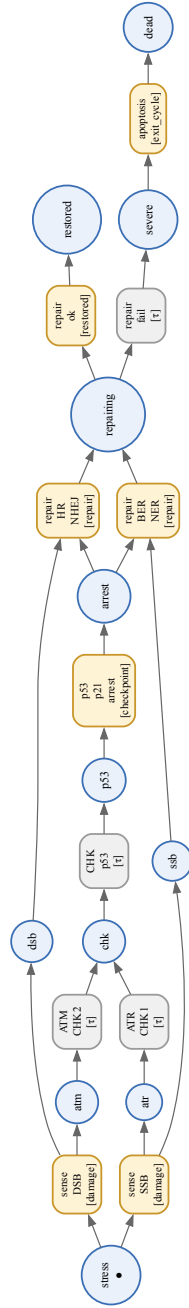


Fig. 3. Detailed animal/human GIM Petri net. Circles are places, rounded boxes are transitions, brackets show the base/coarse observable labels, and gray τ transitions are internal.

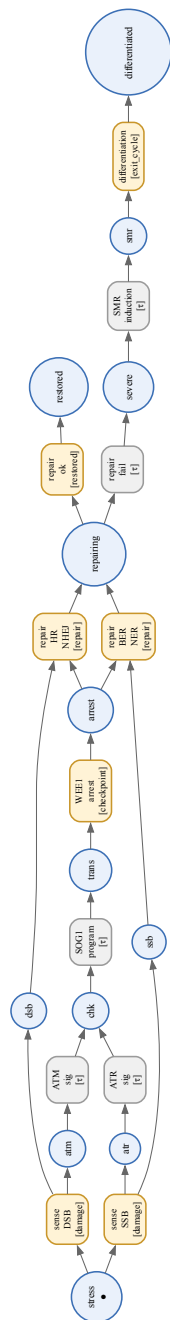


Fig. 4. Detailed *Arabidopsis thaliana* GIM Petri net, using the same graphical conventions as Fig. 3. The terminal transition is a combined outcome in this encoding and does not distinguish differentiation from endoreduplication.

Data and code availability

Code, curated definitions, hash-verified public models, normalized response files, metadata, and generated results are available under the MIT License at <https://github.com/cero1979/BisimulationMol>. The Journal directory supplies the accepted article and supplementary sources. The repository includes the executed notebook, exact continuation outputs, and execution protocol. The repository documents formal/software verification, interface definitions, figure regeneration, tests, and notebook execution. Machine-readable GIM and HPN results are provided as `results/gim_interface_sensitivity.csv` and `results/hpn_dream_formal_data_validation.csv`. No confidential or participant-level data were used.

Declarations

Funding. No specific funding was received. **Competing interests.** The author declares no competing interests. **Ethics.** Only computational models and published aggregate data were used; no human participants, samples, or live animals were involved. **Author contributions.** Carlos Ramirez Ovalle performed the conceptualization, methodology, software, formal analysis, validation, visualization, drafting, and revision.

References

1. Calzone L, Tournier L, Fourquet S, Thieffry D, Zhivotovsky B, Barillot E, Zinovyev A, Mathematical modelling of cell-fate decision in response to death receptor engagement, *PLoS Computational Biology* **6**(3):e1000702, 2010. doi:10.1371/journal.pcbi.1000702.
2. Chaouiya C, Petri net modelling of biological networks, *Briefings in Bioinformatics* **8**(4):210–219, 2007. doi:10.1093/bib/bbm029.
3. Li C, Suzuki S, Ge QW, Nakata M, Matsuno H, Miyano S, Structural modeling and analysis of signaling pathways based on Petri nets, *Journal of Bioinformatics and Computational Biology* **4**(5):1119–1140, 2006. doi:10.1142/S021972000600234X.
4. Fisher J, Henzinger TA, Executable cell biology, *Nature Biotechnology* **25**(11):1239–1249, 2007. doi:10.1038/nbt1356.
5. Szawulak B, Formanowicz P, Graphlets in comparison of Petri net-based models of biological systems, *Scientific Reports* **12**:20942, 2022. doi:10.1038/s41598-022-24535-5.
6. Milner R, *Communication and Concurrency*, Prentice Hall, New York, 1989. ISBN 978-0-13-115007-2.
7. Sangiorgi D, *Introduction to Bisimulation and Coinduction*, Cambridge University Press, Cambridge, 2011. ISBN 978-1-107-00363-7. doi:10.1017/CBO9780511777110.
8. Jackson SP, Bartek J, The DNA-damage response in human biology and disease, *Nature* **461**(7267):1071–1078, 2009. doi:10.1038/nature08467.
9. Blackford AN, Jackson SP, ATM, ATR, and DNA-PK: The trinity at the heart of the DNA damage response, *Molecular Cell* **66**(6):801–817, 2017. doi:10.1016/j.molcel.2017.05.015.
10. Yoshiyama KO, Sakaguchi K, Kimura S, Dna damage response in plants: conserved

- and variable response compared to animals, *Biology* **2**(4):1338–1356, 2013. doi:10.3390/biology2041338.
11. De Schutter K, Joubès J, Cools T, Verkest A, Corellou F, Babychuk E, Van Der Schueren E, Beeckman T, Kushnir S, Inzé D, De Veylder L, Arabidopsis WEE1 kinase controls cell cycle arrest in response to activation of the DNA integrity checkpoint, *The Plant Cell* **19**(1):211–225, 2007. doi:10.1105/tpc.106.045047.
 12. Yi D, Kamei CLA, Cools T, Vanderauwera S, Takahashi N, Okushima Y, Eekhout T, Yoshiyama KO, *et al.*, The Arabidopsis SIAMESE-RELATED cyclin-dependent kinase inhibitors SMR5 and SMR7 regulate the DNA damage checkpoint in response to reactive oxygen species, *The Plant Cell* **26**(1):296–309, 2014. doi:10.1105/tpc.113.118943.
 13. Adachi S, Minamisawa K, Okushima Y, Inagaki S, Yoshiyama K, Kondou Y, Kaminuma E, Kawashima M, Toyoda T, Matsui M, Kurihara D, Matsunaga S, Umeda M, Programmed induction of endoreduplication by DNA double-strand breaks in Arabidopsis, *Proceedings of the National Academy of Sciences* **108**(24):10004–10009, 2011. doi:10.1073/pnas.1103584108.
 14. Hill SM, Heiser LM, Cokelaer T, Unger M, Nesser NK, Carlin DE, *et al.*, Inferring causal molecular networks: Empirical assessment through a community-based effort, *Nature Methods* **13**(4):310–318, 2016. doi:10.1038/nmeth.3773.
 15. Razzaq M, Paulevé L, Siegel A, Saez-Rodriguez J, Bourdon J, Guziolowski C, Computational discovery of dynamic cell line specific boolean networks from multiplex time-course data, *PLOS Computational Biology* **14**(10):e1006538, 2018. doi:10.1371/journal.pcbi.1006538.
 16. Ogita N, Okushima Y, Tokizawa M, Yamamoto YY, Tanaka M, Seki M, Makita Y, Matsui M, Okamoto-Yoshiyama K, *et al.*, Identifying the target genes of SUPPRESSOR OF GAMMA RESPONSE 1, a master transcription factor controlling DNA damage response in Arabidopsis, *The Plant Journal* **94**(3):439–453, 2018. doi:10.1111/tpj.13866.
 17. Manova V, Gruszka D, DNA damage and repair in plants—from models to crops, *Frontiers in Plant Science* **6**:885, 2015. doi:10.3389/fpls.2015.00885.
 18. Fulcher N, Sablowski R, Hypersensitivity to DNA damage in plant stem cell niches, *Proceedings of the National Academy of Sciences* **106**(49):20984–20988, 2009. doi:10.1073/pnas.0909218106.
 19. Murata T, Petri nets: Properties, analysis and applications, *Proceedings of the IEEE* **77**(4):541–580, 1989. doi:10.1109/5.24143.
 20. Quimbaya M, Vandepoele K, Raspé E, Matthijs M, Dhondt S, Beemster GTS, Berx G, De Veylder L, Identification of putative cancer genes through data integration and comparative genomics between plants and humans, *Cellular and Molecular Life Sciences* **69**(12):2041–2055, 2012. doi:10.1007/s00018-011-0909-x.
 21. Clavijo-Buriticá DC, Sosa CC, Cárdenas-Heredia R, Mosquera AJ, Álvarez A, Medina J, Quimbaya M, Use of Arabidopsis thaliana as a model to understand specific carcinogenic events: Comparison of the molecular machinery associated with cancer-hallmarks in plants and humans, *Heliyon* **9**(4):e15367, 2023. doi:10.1016/j.heliyon.2023.e15367.
 22. Hanahan D, Hallmarks of cancer: new dimensions, *Cancer Discovery* **12**(1):31–46, 2022. doi:10.1158/2159-8290.CD-21-1059.
 23. Naldi A, Hernandez C, Abou-Jaoudé W, Monteiro PT, Chaouiya C, Thieffry D, Logical modeling and analysis of cellular regulatory networks with GINsim 3.0, *Frontiers in Physiology* **9**:646, 2018. doi:10.3389/fphys.2018.00646.
 24. Chaouiya C, Béranguier D, Keating SM, Naldi A, van Iersel MP, Rodriguez N, *et al.*, SBML qualitative models: a model representation format and infrastructure to fos-

- ter interactions between qualitative modelling formalisms and tools, *BMC Systems Biology* **7**:135, 2013. doi:10.1186/1752-0509-7-135.
25. Traynard P, Fauré A, Fages F, Thieffry D, Logical model specification aided by model-checking techniques: Application to the mammalian cell cycle regulation, *Bioinformatics* **32**(17):i772–i780, 2016. doi:10.1093/bioinformatics/btw457.
 26. Abou-Jaoudé W, Ouattara DA, Kaufman M, From structure to dynamics: Frequency tuning in the p53–mdm2 network, *Journal of Theoretical Biology* **258**(4):561–577, 2009. doi:10.1016/j.jtbi.2009.02.005.
 27. Radom M, Rybarczyk A, Szawulak B, Andrzejewski H, Chabelski P, Kozak A, Formanowicz P, Holmes: A graphical tool for development, simulation and analysis of Petri net based models of complex biological systems, *Bioinformatics* **33**(23):3822–3823, 2017. doi:10.1093/bioinformatics/btx492.
 28. Bunte O, Groote JF, Keiren JJA, Laveaux M, Neele T, de Vink EP, Wesselink W, Wijes A, Willemse TAC, The mCRL2 toolset for analysing concurrent systems: Improvements in expressivity and usability, *Tools and Algorithms for the Construction and Analysis of Systems (TACAS 2019)*, Lecture Notes in Computer Science, Vol. 11428, Springer, pp. 21–39, 2019. doi:10.1007/978-3-030-17465-1_2.